

As soon as the user has selected the drop down value, it should be reflected as the text value of the same. The code given in Figure 7 makes this conversion.

We want the conversion to happen as soon as the button is pressed, but we also need to make sure that the users have provided all the necessary inputs, i.e., they should have selected at least one of the conversion modes, have selected the 'From' unit from the first drop down box, provided some input value in the input text box and, finally, given the conversion unit for the 'To' drop down box. Listed below are the validations for each case.

- Checking the checkbox status (Figure 8).
- Further checking whether or not the user has provided some input value (Figure 9).

If everything seems okay, the output value should be calculated (Figure 10).

Now we are done with the block editor too. Let us move to downloading and installing the app on the phone to check how it works.

Packaging and testing

To test the app, you need to get it on your phone. First, you have to download it to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

1. On the top row, clicking on the 'Build' button will show you the option to download the APK to your computer.
2. You can watch the progress of the download and, after it is successfully done, the application will be placed in the *Download* folder of your directory or the preferred location you have set for it.
3. Now you need to get this APK file to your mobile phone either via Bluetooth or via USB cable. Once you have placed the APK file on your SD card, you need to install it. Follow the on-screen instructions to install it. You might get some notification or warning like, 'Install from untrusted source'. Allow this from the *Settings* and after the successful installation, you will see the icon of



Figure 7: Block Editor Image 4



Figure 8: Block Editor Image 5



Figure 9: Block Editor Image 6



Figure 10: Block Editor Image 7

your application in the menu of your mobile. This will be the default icon, and I will tell you how it can be changed as we move ahead.

As you can see, we have put in place the logic to convert units of weight. Now you need to think about the logic for the remaining conversions.

I hope your application is working exactly according to the specs you have given. Now, depending upon your usability and customisation requirements, you can change various things like the image, sound and behaviour too.

Debugging the application

We have just created the prototype of the application with very basic functionality, but what else might the user be interested in? There are various

use cases that require serious thought so as not to annoy the user. Consider the following use cases:

1. How will you code it to pick the conversion type and then apply the multiplication factor?
2. If there is more than one value in the drop down menu, how will the multiplication factor be decided?
3. What if the user doesn't select the drop down value and it remains with the default 'Select' text?

These are some of the scenarios that might occur and users will be pretty happy seeing these addressed.

Think over all these scenarios—how can you integrate them into the application? Do ask me if you fail to accomplish any of the above use cases.

You have successfully built another useful Android app for yourself. Happy inventing! 

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